

House Price Predictor

A Professional Machine Learning Solution for Real Estate Valuation

This repository contains a comprehensive House Price Prediction system, utilizing advanced regression techniques and a streamlined user interface to provide accurate property valuations.



Streamlit Web User Interface

The application features a clean, responsive web interface built using Streamlit. It organizes inputs into logical sections—General Information, Room & Floor Sizes, Rooms/Bathrooms/Garage, and Additional Features—with helpful tooltips and real-time metric updates.



House Price Prediction App

Please provide the details of the property below, and the AI will estimate its market value.

1. General & Building Information

Building Style / Type 	Material & Finish Quality 	Year Built 
20	5	2000
Total Land Area (sq ft) 	Overall House Condition 	Year Remodeled 
10000	5	2000

2. Room & Floor Sizes

Finished Basement Area (sq ft)	1st Floor Area (sq ft)	Wooden Deck Area (sq ft)
500.00	850.00	0.00



Dataset Information

- **Source Dataset:-**
https://www.kaggle.com/datasets/jenilhareshbhaighori/house-price-prediction/data?select=house_cleaned.csv
- **Target Variable (y):** `SalePrice` (Final selling price of the house).
- **Feature Engineering & Preprocessing:**
 - **Identifier Removal:** Dropped the unique identifier column (`Id`) as it adds no predictive value.
 - **Data Integrity:** Checked for missing values, handled data types, and split the dataset using an 80-20 train-test split (`random_state=42`).
 - **Normalization:** Utilized `StandardScaler` where necessary to normalize feature scales (mean = 0, standard deviation = 1) for distance-sensitive algorithms.
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Dataset Feature Dictionary

Column	Meaning
<code>Id</code>	Unique ID number for each house. Usually not useful for prediction because it is just an identifier.
<code>MSSubClass</code>	Type/class of the house based on the building classification. It is a coded value such as 20, 30, 60, etc.
<code>LotArea</code>	Total area of the land/plot on which the house is built.
<code>OverallQual</code>	Overall quality of the house/materials and finish. Higher value = better quality.
<code>OverallCond</code>	Overall condition of the house. Higher value = better condition.
<code>YearBuilt</code>	Year in which the house was originally built.
<code>YearRemodAdd</code>	Year when the house was last remodeled/renovated.

Column	Meaning
BsmtFinSF1	Finished basement area associated with the first type of basement finishing.
BsmtUnfSF	Unfinished basement area.
TotalBsmtSF	Total basement area.
WoodDeckSF	Area of the outdoor wooden deck.
OpenPorchSF	Area of the open porch.
EnclosedPorch	Area of enclosed porch.
3SsnPorch	Area of a three-season porch.
ScreenPorch	Area of a screened porch.
PoolArea	Area of the swimming pool.
MiscVal	Value of miscellaneous features that aren't covered by the other columns.
MoSold	Month in which the house was sold.
YrSold	Year in which the house was sold.
SalePrice	Final selling price of the house. This is normally your target (y) for house-price prediction.



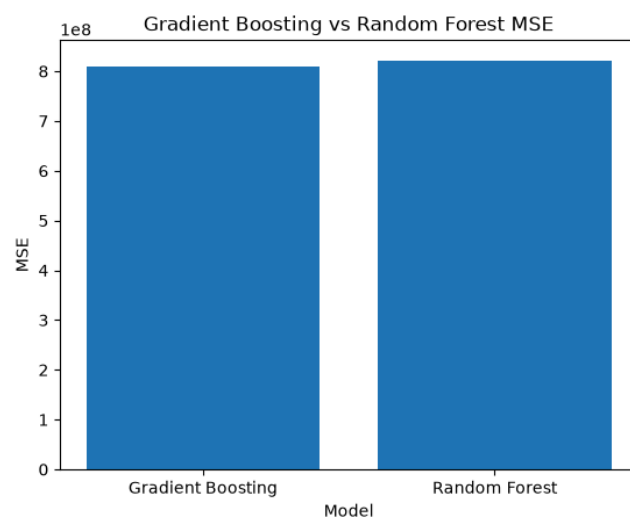
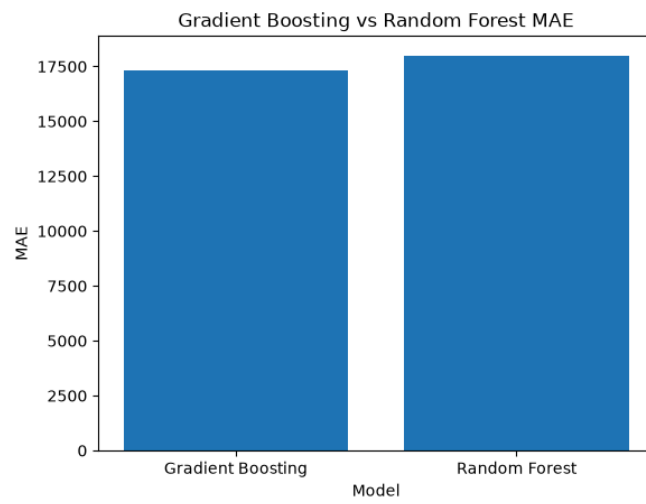
Machine Learning Model & Algorithm Selection

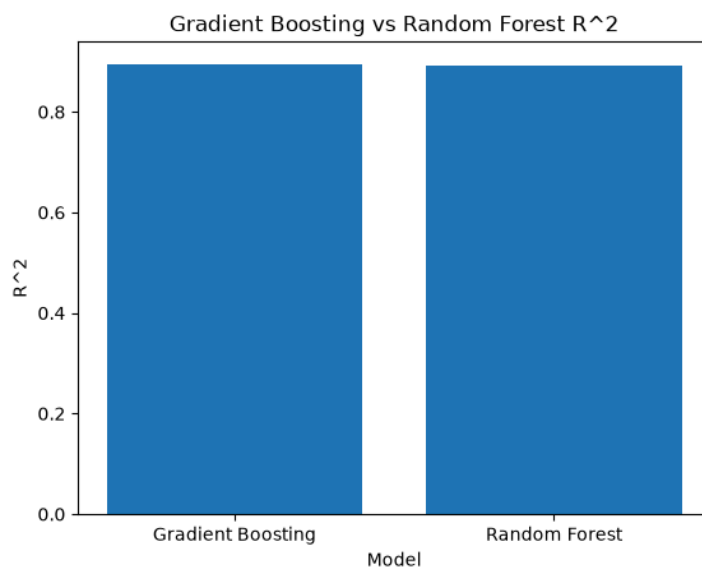
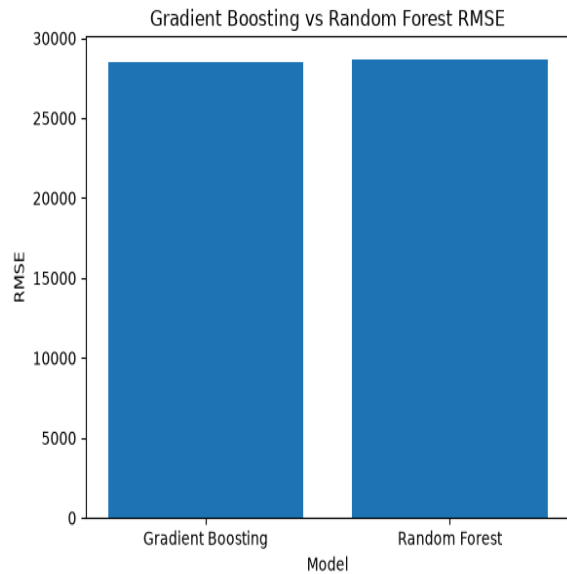
To achieve optimal predictive accuracy, three distinct machine learning regression models were trained, evaluated, and compared:

Model	Technique	Key Characteristics
Linear Regression	Baseline	Understands fundamental linear relationships between features.
Random Forest	Bagging Ensemble	Uses multiple decision trees (n_estimators=300) to capture non-linear patterns.
Gradient Boosting	Boosting Ensemble	Selected Production Model. Sequentially corrects errors from previous trees.

Why Gradient Boosting?

- **Sequential Error Correction:** It builds trees sequentially, where each new tree specifically targets and minimizes the residual errors of the previous trees.
- **Superior Performance on Tabular Data:** Real estate pricing involves complex, non-linear interactions (e.g., how age, square footage, and overall quality compound together). Gradient Boosting models excel at capturing these nuances without requiring heavy manual feature scaling.
- **Hyperparameters:** `n_estimators=300`, `learning_rate=0.05`, `max_depth=3`, `random_state=42`.





Evaluation Metrics & Model Comparison

Models were evaluated using standard regression metrics to ensure reliability:

- **MAE (Mean Absolute Error):** Measures the average magnitude of absolute prediction errors.
- **MSE (Mean Squared Error):** Squares errors, heavily penalizing larger outlier mistakes.
- **RMSE (Root Mean Squared Error):** Interpretable error metric in the exact same units as SalePrice (\$).

- **R² Score (Coefficient of Determination):** Represents the proportion of variance in the target variable explained by the features (closer to 1.0 is better).

All nurmiacal data:-

- R2: 0.894401570332499
- MAE : 17295.32
- MSE : 809974402.52
- RMSE : 28460.05
- R2 : 0.89

Gradient Boosting consistently yielded the highest R² score and lowest error margins compared to baseline and standard random forest configurations.

Tech Stack & Libraries

- **Python:** Core Programming Language
- **Pandas & NumPy:** Data Manipulation & Numerical Operations
- **Scikit-Learn:** Model Training, Metrics, and Scalers
- **Matplotlib:** Data Visualization & Model Comparison Charts
- **Joblib:** Model Serialization
- **Streamlit:** Interactive Web UI Framework

Installation & Local Execution

Follow these steps to run the application on your local machine:

1. **Clone the repository:**

```
git clone <span type="placeholder"
placeholder-type="person"></span>/house-price-predictor
```

2. **Install required dependencies:**

```
pip install -r requirements.txt
```

3. **Run the Streamlit application:**

```
streamlit run CODE/house_price_predictor_streamlit_ui.py
```

Deployment

This project is deployment-ready for **Streamlit Community Cloud**:

1. Push your repository files to GitHub.
 2. Link your repository on the Streamlit Cloud dashboard.
 3. Set the main file path to
`CODE/house_price_predictor_streamlit_ui.py` and deploy!
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